

Experiment 6

Problem Statement:

Develop a structure based social media analytics model for any business. Build models for community detection and influence analysis using libraries like NetworkX, Gephi, pr Neo4j. Analyze social media graphs and connections using APIs like LinkedIn API or Twitter API v2

Theory:

Social media analytics has become a critical tool for businesses to understand their audience, measure engagement, and identify key influencers. A structure-based analytics model leverages network science to detect communities and measure influence within a social media ecosystem. This document outlines the methodology and tools required for implementing such a model.

The key objectives of this model include community detection, influence analysis, and graph-based social media analysis. Community detection helps identify clusters or groups of users with similar interests and interactions, while influence analysis measures the influence of key users based on their connectivity and engagement. The overall goal is to analyze the structure of social media networks to uncover actionable insights.

For implementation, various tools and technologies are utilized, including NetworkX, Gephi, and Neo4j. APIs such as Twitter API v2 and LinkedIn API are used to extract data, while Neo4j and ArangoDB serve as graph databases for storing and querying network structures. Visualization tools like Gephi, Cytoscape, and Matplotlib help interpret and present data effectively.

Data collection is essential for analysis, with sources including Twitter API v2 for follower-friend relationships, mentions, and retweets, and LinkedIn API for professional networks and endorsements. Other sources such as Reddit and Facebook Graph API can also be leveraged. Once collected, data is preprocessed by removing duplicates, converting it into structured formats like GraphML or JSON, and extracting key attributes such as user ID, engagement metrics, and timestamps.

Community detection methods include the Louvain Method, which identifies clusters within a social graph by maximizing modularity, the Girvan-Newman Algorithm, which detects hierarchical community structures using edge betweenness, and the Label Propagation Algorithm, which dynamically assigns labels to nodes for large-scale networks. These techniques can be implemented using NetworkX in Python.

Influence analysis involves centrality measures such as Degree Centrality, which assesses direct connections, Betweenness Centrality, which evaluates brokerage potential, Eigenvector Centrality, which identifies influential users based on their neighbors' influence, and PageRank,

an algorithm adapted for ranking social influencers. These measures help businesses identify key players in a social network who can drive engagement and brand visibility.

Social media graph analysis further examines network properties such as Density, which measures how interconnected a network is, the Clustering Coefficient, which determines the likelihood of forming tight communities, and the Average Path Length, which assesses information flow efficiency. Visualization tools like Gephi allow for effective representation of these insights using layouts like ForceAtlas2 and modularity detection techniques.

A case study illustrating business applications of this model involves identifying key influencers in a product campaign. By extracting Twitter mentions and retweets related to a product hashtag, a social graph can be built with users as nodes and interactions as edges. PageRank helps rank users based on influence, while the Louvain method segments user communities. This approach allows businesses to identify key influencers for targeted marketing outreach.

In conclusion, a structure-based social media analytics model enables businesses to gain deep insights into user interactions, community structures, and influential figures. Leveraging tools like NetworkX, Neo4j, and Gephi helps optimize social media strategies, improve engagement, and enhance marketing efforts. Future advancements can integrate AI-driven sentiment analysis and predictive modeling for even more precise insights.